



មហាវិទ្យាល័យវិស្វកម្ម
FACULTY OF ENGINEERING

Computer Programming

Lecture 7 Control Statement - Loops

Chhoeum Vantha, Ph.D.
Telecom & Electronic Engineering

Contents

- Introduction to Loop
- The while Loop
- The do-while Loop

Introduction to Loop

- A loop can be used to tell a program to execute statements repeatedly.

```
100 times { cout << "Welcome to C++!\n";  
           cout << "Welcome to C++!\n";  
           ...  
           cout << "Welcome to C++!\n";
```

```
int count = 0;  
while (count < 100)  
{  
    cout << "Welcome to C++!\n";  
    count++;  
}
```

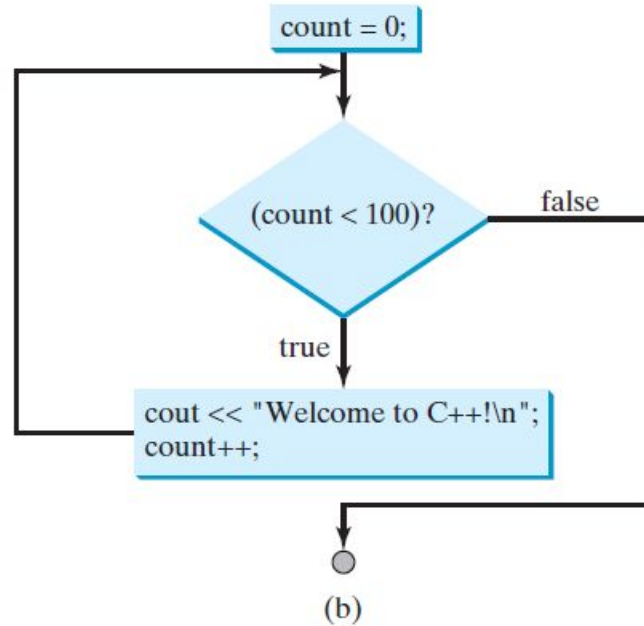
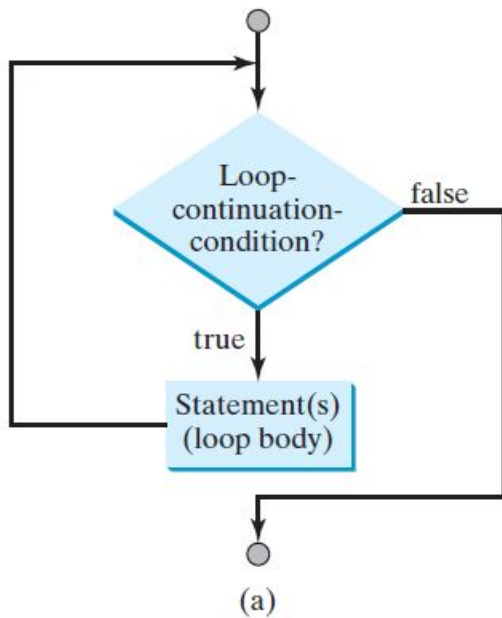
The while Loop

- A while loop executes statements repeatedly while the condition is true.
- The syntax for the while loop is

```
<initialization>
while (<test>)
{
    <statement1>;
    ...
    <statementN>;
    <update>
}
```

```
int count = 0;
while (count < 100)
{
    cout << "Welcome to C++!\n";
    count++;
}
```

The while Loop



```
int count = 0;
while (count < 100)
{
    cout << "Welcome to C++!\\n";
    count++;
}
```

← loop-continuation-condition

} loop body

The while Loop

- The loop-continuation-condition must always appear inside the **parentheses**.
- The **braces enclosing** the loop body can be **omitted** only if the loop body contains **one statement or none**.

```
int sum = 0, i = 1;
while (i < 10)
{
    sum = sum + i;
    i++;
}
```

The while Loop

Note

- Make sure that the loop-continuation-condition eventually becomes **false** so that the loop will **terminate**.
- A **common programming error** involves **infinite loops** (i.e., the loop runs forever).
- If your program takes an **unusually long** time to run and does not stop, it may have an **infinite loop**.

The while loop - Assignment Operators

- Examples of other assignment operators include:

c += 3 (c = c + 3)

d -= 4 (d = d - 4)

e *= 5 (e = e * 5)

f /= 3 (f = f / 3)

g %= 9 (g = g % 9)

The while loop - Example

- Display Numbers from 1 to 5

```
// C++ Program to print numbers from 1 to 5

#include <iostream>

using namespace std;

int main() {
    int i = 1;

    // while loop from 1 to 5
    while (i <= 5) {
        cout << i << " ";
        ++i;
    }

    return 0;
}
```

Output|

1 2 3 4 5

The while loop - Example

- Display Numbers from 1 to 5:

How does the program work?

Iteration	Variable	$i \leq 5$	Action
1st	<code>i = 1</code>	true	1 is printed and <code>i</code> is increased to 2.
2nd	<code>i = 2</code>	true	2 is printed and <code>i</code> is increased to 3.
3rd	<code>i = 3</code>	true	3 is printed and <code>i</code> is increased to 4.
4th	<code>i = 4</code>	true	4 is printed and <code>i</code> is increased to 5.
5th	<code>i = 5</code>	true	5 is printed and <code>i</code> is increased to 6.
6th	<code>i = 6</code>	false	The loop is terminated

The do-while Loop

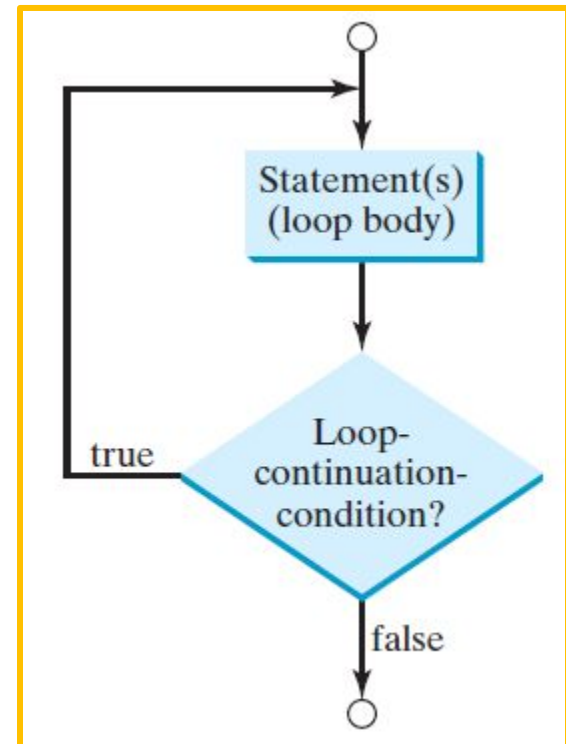
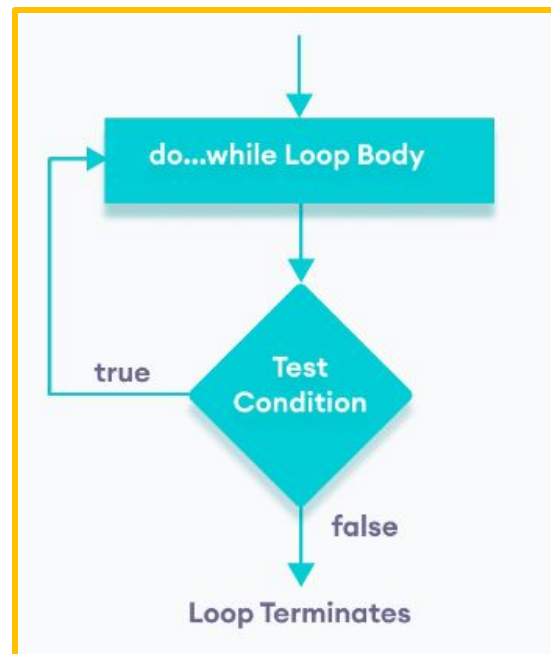
A do-while loop is the same as a while loop except that it executes the loop body first and then checks the loop continuation condition.

The do-while loop is a variation of the while loop.
Its syntax is as follows:

```
do
{
    // Loop body;
    Statement(s);
} while (loop-continuation-condition);
```

The do-while Loop

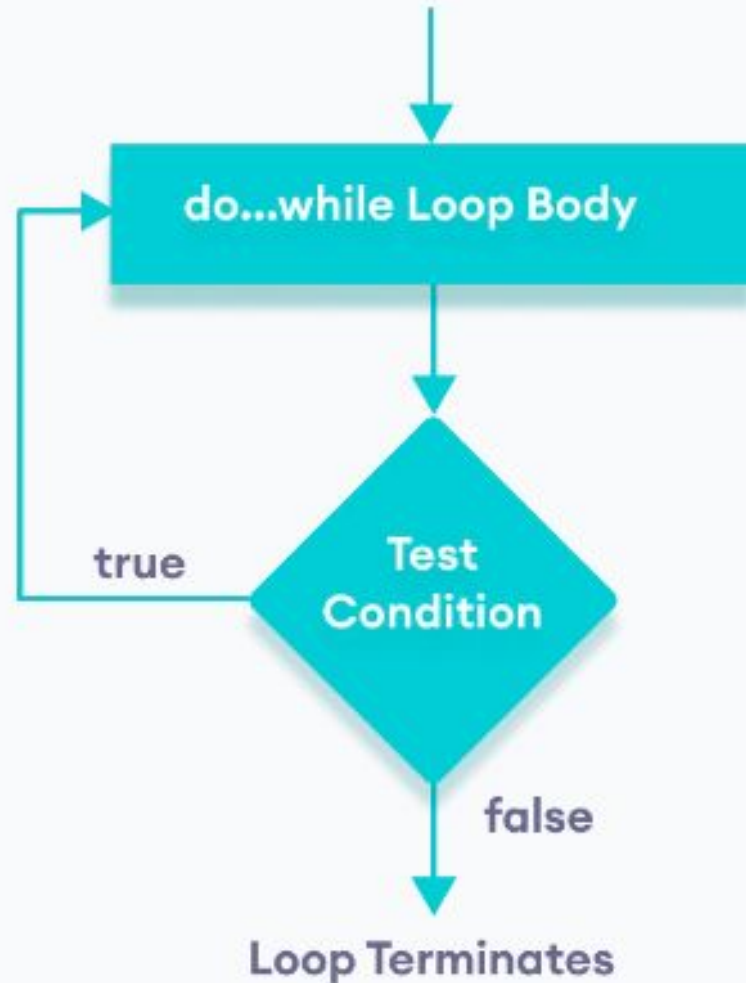
The do-while loop executes the loop body first, then checks the loop-continuation-condition to determine whether to continue or terminate the loop.



The do-while loop

- Syntax: *do statement*

```
do {  
    // body of loop;  
}  
while (condition);
```



The do-while loop

- Example: Display Numbers from 1 to 5

```
// C++ Program to print numbers from 1 to 5

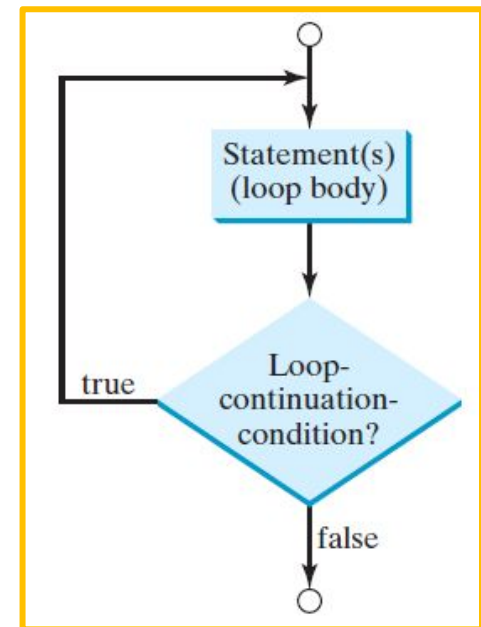
#include <iostream>

using namespace std;

int main() {
    int i = 1;

    // do...while loop from 1 to 5
    do {
        cout << i << " ";
        ++i;
    }
    while (i <= 5);

    return 0;
}
```



Output

1 2 3 4 5

The do-while loop

- Example: Display Numbers from 1 to 5: How it work?

Iteration	Variable	$i \leq 5$	Action
	<code>i = 1</code>	not checked	<code>1</code> is printed and <code>i</code> is increased to 2
1st	<code>i = 2</code>	true	<code>2</code> is printed and <code>i</code> is increased to 3
2nd	<code>i = 3</code>	true	<code>3</code> is printed and <code>i</code> is increased to 4
3rd	<code>i = 4</code>	true	<code>4</code> is printed and <code>i</code> is increased to 5
4th	<code>i = 5</code>	true	<code>5</code> is printed and <code>i</code> is increased to 6
5th	<code>i = 6</code>	false	The loop is terminated

The do-while loop

- Example: How is the result of the code below?

```
#include <iostream>
using namespace std;

int main () {
    // Local variable declaration:
    int a = 10;

    // do loop execution
    do {
        cout << "value of a: " << a << endl;
        a = a + 1;
    } while( a < 20 );

    return 0;
}
```


The do-while loop

- Example: How is the result of the code below?

```
int main()
{
    // initialization expression
    int i = 1;

    // test expression
    do
    {
        cout<<"Hello World\n";

        // update expression
        i=i+1;
    }
    while(i<=5);
    return 0;
}
```

The do-while loop

- Example: How is the result of the code below?

```
int main()
{
    int i=1;
    do
    {
        cout<<i<<endl;
        i=i+1;
    }
    while(i<=10);
    return 0;
}
```

W7 - Practice

Exercise 1.

Write a program in C++ to print the multiplication table.

Example: Input the number of multiplication: 5

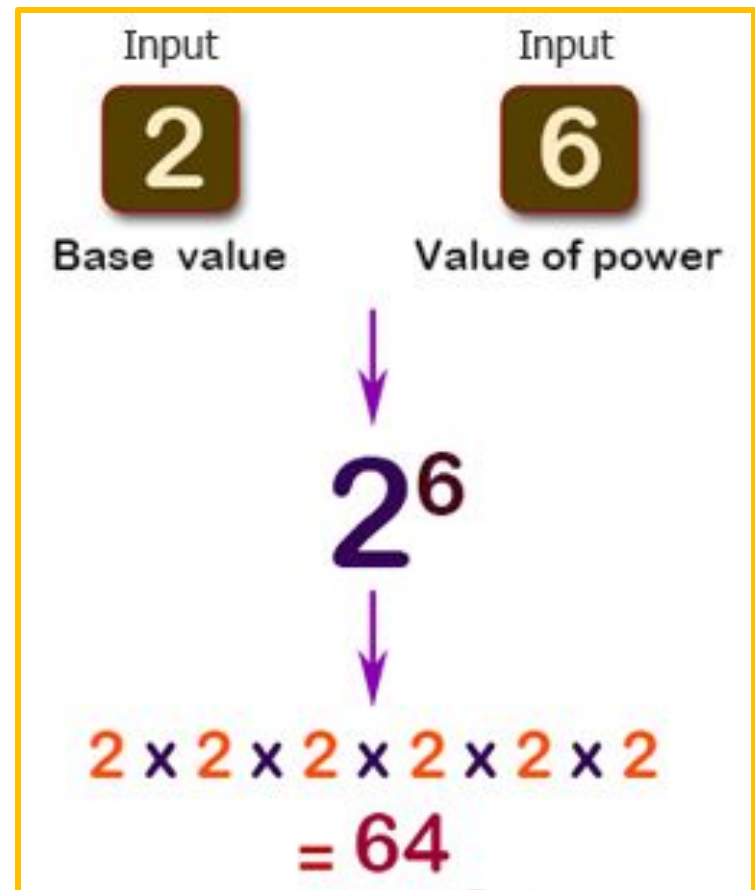
- Here is a sample run:

```
5 x 1  = 5
5 x 2  = 10
5 x 3  = 15
5 x 4  = 20
5 x 5  = 25
5 x 6  = 30
5 x 7  = 35
5 x 8  = 40
5 x 9  = 45
5 x 10 = 50
```

Exercise 2.

Write a program in C++ to find power of any number using while loop:

- Here is a sample run:



Thanks!